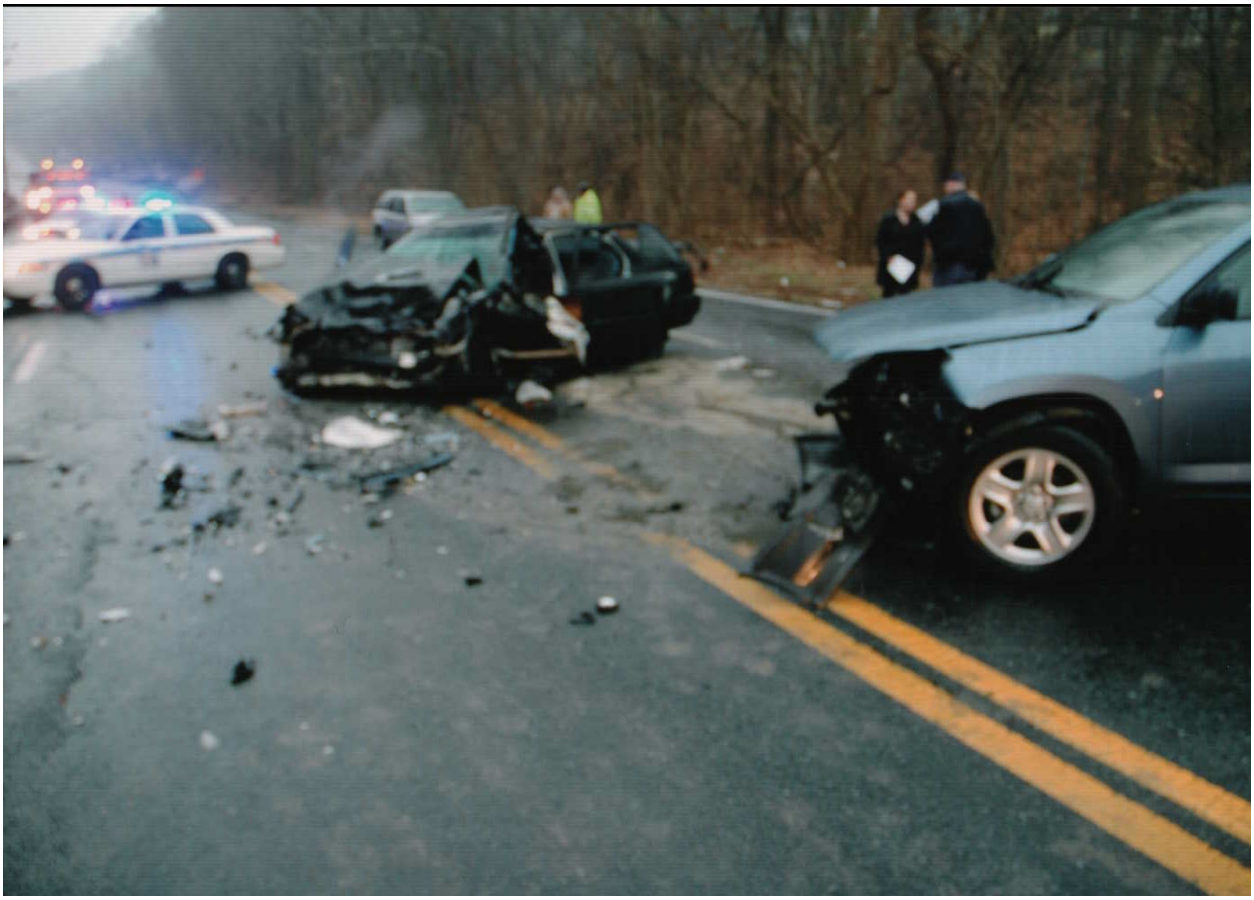
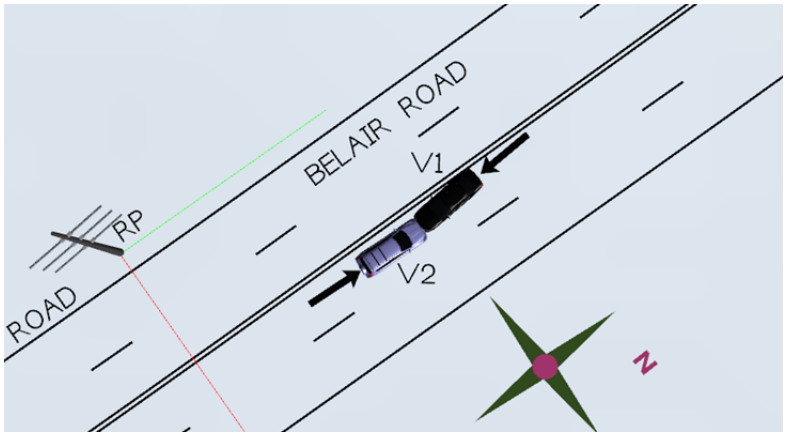
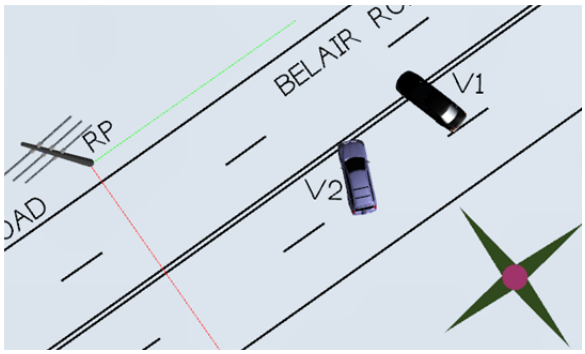
Honda hits Toyota RAV4 Head-on Project with Toyota EDR data



A 1989 Honda Accord appears to have crossed the centerline and impacted a 2011 Toyota RAV4 in a nearly head-on, full overlap collision. The vehicles rebound from one another and come to a stop very close to the point of impact. Based on the scene diagram, the RAV4 CG was pushed back approximately 5 feet and rotated 60 degrees. The Honda rotated 90 degrees and the CG was pushed back approximately 2 feet. The road for the Honda had just curved to the right slightly and it appears the Honda simply kept going straight, crossing the centerline. Road conditions were a mixture of rain, sleet, and snow on the roadway. The speed limit was 50 mph. The wet asphalt drag factor was 0.55g. The Honda Accord weighed 2705 lbs including driver, the Toyota RAV4 weighted 3507 lbs. including the driver. The Toyota driver was a small statured female and was pronounced dead at the scene. The Honda is too old to have an event data recorder, but the 2011 Toyota RAV4 has one. You were asked to image the Toyota module and provide your insights into the crash.



Area of Impact



Final Rest



RAV 4 Post Impact



Honda Post Impact

1. Review the CDR Report. How many events are recorded? Are all events complete and valid recordings?

System Status at Time of Retrieval

ECU Part Number	89170-42380
ECU Generation	04EDR
Recording Status, All Pages	Complete
Freeze Signal	ON
Freeze Signal Factor	Front Airbag Deployment Front Pretensioner Deployment
Diagnostic Trouble Codes Exist	No
Time from Previous Pre Crash TRG (msec)	16381 or greater
Latest Pre-Crash Page	0
Contains Unlinked Pre-Crash Data	No

Event Record Summary at Retrieval

Events Recorded	TRG Count	Crash Type	Time (msec)	Pre-Crash & DTC Data Recording Status	Event & Crash Pulse Data Recording Status
Most Recent Event	2	Front/Rear Crash	0	Complete (Page 0)	Complete (Front/Rear Page 0)
1st Prior Event	1	Side Crash	-1	Complete (Page 0)	Complete (Side Page 0)

2. Of the events recorded, how many are from your crash?

3. Using only the RAV4 pre-crash speed data, use the speed at impact worksheet to determine a range of speeds at impact. To do this, you must decide how old data point you use could be.

Pre-Crash Data, -5 to 0 seconds (Most Recent Event, TRG 2)

Time (sec)	-4.4	-3.4	-2.4	-1.4	-0.4	0 (TRG)
Vehicle Speed (MPH [km/h])	48.5 [78]	49.7 [80]	49.7 [80]	49.7 [80]	42.3 [68]	32.3 [52]
Brake Switch	OFF	OFF	OFF	ON	ON	ON
Accelerator Rate (V)	0.98	0.86	0.78	0.78	0.78	0.78
Engine RPM (RPM)	1,600	1,600	1,600	1,600	1,200	800

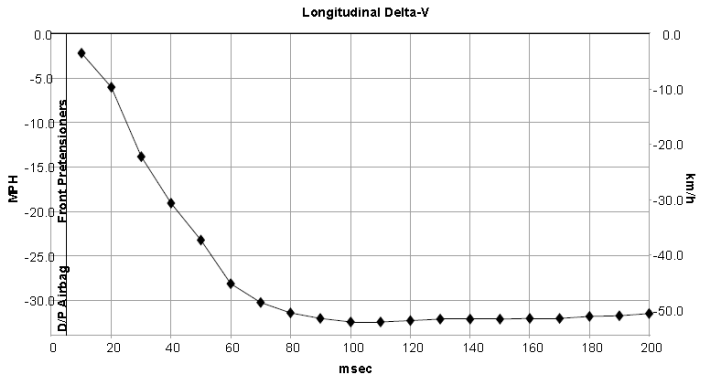
You must choose whether to start from the last data point of 32.3 or the next to last data point of 42.3 – if you are not sure which is best do it both ways.

	For 0 (TRG)		For "-0.4"	
	MIN	MAX	MIN	MAX
Starting Data Point "0 (TRG)" or "-0.4"	32.3	32.3	42.3	42.3
Additional Braking Prior to Impact (0.4 sec x 12 mph/sec)	- (.4)	+0.0	- (.9)	(.5 sec)
Slip Adjustment (+5% due to underreporting in braking)	+	+	+0.0?	+0.0?
Speedometer Error +/-4%	-___	+___	- ___	+___
Range of possible speeds at impact				

4. Based on the Longitudinal Delta V of the RAV4, and making the assumption that it was a nearly an inline central collision, what is the Delta V of the Honda? Newtons laws say crash forces are equal and opposite.

Longitudinal Crash Pulse (Most Recent Event, TRG 2 - table 1 of 2)

Recording Status, Time Series Data	Complete
Max Longitudinal Delta-V (MPH [km/h])	-32.4 [-52.1]



$\Delta V_2 = -\Delta V_1 \frac{W1}{W2}$ The sign of the answer is from a SCENE based co-ordinate system where the RAV4 direction of approach is positive!

ΔV_2

$\Delta V_2 =$ (sign is from Scene based co-ordinate system – RAV4 approach is +)

5. Combining the RAV4 and Honda Delta V's, and assuming 10% restitution, what is the nominal closing speed of the two vehicles? What is the range of possible closing speeds?

Closing Speed = $(\frac{1}{1+RESTITUITION})(\Delta V_2 - \Delta V_1)$ =

(Range +/-10%,)

6. What is the nominal Speed at Impact of the Toyota using the RAV4 ΔV and post-crash movement? What is the range considering the accepted error rate in the data?
   Closing speed alone does not get you to a speed at impact – you must know at least one of the precrash or postcrash speeds of one of the two vehicles. The RAV4 was pushed back 5 feet at a 30 degree angle. The equivalent drag factor after the front 2 wheels locked and the vehicle rotated was 0.4g. Departure speed S = V_4 =√30*d*f = mph.

   RAV4 Speed at Impact = V_4*Cos(β) –ΔV_2 Where β is the departure angle. Since the RAV4 was pushed backward you must insure the sign has departure speed canceling Delta V

   RAV4 Speed at Impact = = mph

   The tolerance on the ΔV is +/-10% of ____ or +/- ____ mph, then the range of possible speeds at impact for the RAV 4 would be ____ to _____ mph.

   If there was a tolerance on the drag factor or postcrash travel distance that would increase the range farther.

7. What is the nominal Speed at Impact of the Honda using the ΔV you calculated for it and the Honda post-crash movement?
   The Honda CG was pushed back 2 feet at a 10 degree angle. The equivalent drag factor after the front 2 wheels locked and the vehicle rotated was 0.4g.

   Honda Departure speed S = V_3 =√30*d*f = mph.

   Honda Speed at Impact = V_3*Cos(β) –ΔV_1 = Where β is the departure angle.
   Check sign on departure speed; it must cancel out part of the Delta V reducing impact speed.

   Honda Speed at Impact = **mph**

   (+/-10% ΔV, Range _____ to _____)

   Note the sign comes from the scene based system where the RAV4 direction of approach was positive.

   Memo: RAV4 speed of ____ plus Honda speed _____ = ____ closing speed, which is less than the _____ calculated earlier using 10% restitution. Restitution may be higher than 10%.

8. Considering both the speed data and the Delta V in the EDR, what is the range of possible speeds at impact of the RAV 4? (Combine the ranges)

RAV 4 Range From EDR Speed Data: ____ to ____ mph |----------|

RAV 4 Range from EDR Delta V: ____ to ____ mph |----------|

RAV 4 Combined EDR Range: ____ **to ____** mph |----|

(You would also combine any range from the crush damage and momentum , if available).

9. Where or when or how was the crime or civil negligence committed?

10. Is there evidence in the EDR that the Toyota driver was awake and alert and attempting to avoid the crash or minimize its severity? When is the first reaction recorded in the EDR?
    Accel pedal release between ____ and ____
    Brake pedal application between ____ and ____

THE INSTRUCTOR MAY DIRECT BASIC STUDENTS TO STOP HERE (ADVANCED STUDENTS PROCEED)

11. Another crash analyst was assigned to check your work and he thinks you got it all wrong, that the vehicles were originally traveling in the opposite direction you thought they were, that the Honda was in its lane and the Toyota crossed the centerline, and that after contact the two vehicles rotated about one another and came to rest on the far side of one another. Due to the wet roadway there were very few visible marks on the road, and most of those got obscured by the rain and snow. That would mean the Honda hit, spun, and the center of gravity traveled 16+2 = 18 feet forward, and that the Toyota hit, spun, and the CG traveled 16+5 = 21 feet. Under this scenario, what is the nominal and range of possible speeds at impact for the Toyota RAV4?

    RAV4 Departure speed S = V_4 =√30*d*f = mph.

    RAV4 Speed at Impact = V_4*Cos(β) –ΔV_2 = mph

    (Range +/-10% of Delta V or +/-3.2mph = ____ to ____ mph)

12. Under this scenario, what is the nominal and range of possible speeds at impact for the Honda?

    Honda Departure speed S = V_4 =√30*d*f = mph.

    Honda Speed at Impact = V_4*Cos(β) –ΔV_2 = mph

    (Range +/-10% of Delta V or +/-___mph = ____ to ____ mph)

13. Under this scenario, what is the closing speed (nominal and range)?

    RAV4 ____mph + Honda ____ mph = ____ mph

    Range from low of _____mph to high of _____ mph

14. Compare this to the range of closing speeds you got in #5 from the reported Delta V in the RAV4 of ____ **to ____ mph.** Of course, the Delta V could be understated due to clipping. **Looking at the crush damage in the photos, which do you think is more correct, the 67 mph nominal closing or the 102 mph closing? If it was 102, what would be the percentage of under-reporting in the EDR Delta V in the Toyota? _____** . In addition, you can look at the damage to each vehicle and see if it is generally even across the front or if it is offset to one side. Some offset (left front to left front or right front to right front) would be necessary to create the rotation required for the vehicles to get past one another (they either both have to go clockwise or both go counterclockwise). The front fender forward of the wheel well on the heavier contact side corner that the vehicles are rotating about should have at least some contact damage if the vehicles rotated around one another.

15. If you had crush measurements, could you also use crush energy methods **to check which closing speed is more correct?**

Last revised February 17, 2013 by RR

IMPORTANT NOTICE: Robert Bosch LLC and the manufacturers whose vehicles are accessible using the CDR System urge end users to use the latest production release of the Crash Data Retrieval system software when viewing, printing or exporting any retrieved data from within the CDR program. Using the latest version of the CDR software is the best way to ensure that retrieved data has been translated using the most current information provided by the manufacturers of the vehicles supported by this product.

CDR File Information

User Entered VIN	JTMJF4DV3B5041258
User	Off. Eckstein #3900
Case Number	12-039-1135
EDR Data Imaging Date	02/08/2012
Crash Date	02/08/2012
Filename	JTMJF4DV3B5041258_ACM.CDRX
Saved on	Wednesday, February 8 2012 at 18:52:21
Collected with CDR version	Crash Data Retrieval Tool 4.3
Reported with CDR version	Crash Data Retrieval Tool 10.0.1
EDR Device Type	Airbag Control Module
Event(s) recovered	Front/Rear (1), Side (1)

Comments

No comments entered.

Data Limitations

CDR Record Information:

- Due to limitations of the data recorded by the airbag ECU, such as the resolution, data range, sampling interval, time period of the recording, and the items recorded, the information provided by this data may not be sufficient to capture the entire crash.
- Pre-Crash data is recorded in discrete intervals. Due to different refresh rates within the vehicle's electronics, the data recorded may not be synchronous to each other.
- Airbag ECU data should be used in conjunction with other physical evidence obtained from the vehicle and the surrounding circumstances.
- If the airbags did not deploy or the pretensioners did not operate during an event that meets a specified recording threshold, it is called a Non-Deployment Event. Data from a Non-Deployment Event can be overwritten by a succeeding event that meets the specified recording threshold. If the airbag(s) deploy or the pretensioners are operated, it is called a Deployment Event. Deployment Event data cannot be overwritten or deleted by the airbag ECU following that event.
- If power supply to the airbag ECU is lost during an event, all or part of the data may not be recorded.
- "Diagnostic Trouble Codes" are information about faults when a recording trigger is established. Various diagnostic trouble codes could be set and recorded due to component or system damage during an accident.
- The airbag ECU records only diagnostic information related to the airbag system. It does not record diagnostic information related to other vehicle systems.
- The TaSCAN, Global TechStream, or Intelligent Tester II devices (or any other Toyota genuine diagnostic tool) can be used to obtain detailed information on the diagnostic trouble codes from the airbag system, as well as diagnostic information from other systems. However, in some cases, the diagnostic trouble codes of the airbag system recorded by the airbag ECU when the event occurred may not match the diagnostic trouble codes read out when the diagnostic tool is used.

General Information:

- The data recording specifications of Toyota's airbag ECUs are divided into the following six categories. The specifications for 12EDR are designed to be compatible with NHTSA's 49CFR Part 563 rule.
  - 00EDR / 02EDR / 04EDR / 06EDR / 10EDR / 12EDR
- The airbag ECU records data for all or some of the following accident types: frontal crash, rear crash , side crash, and rollover events. Depending on the installed airbag ECU, data for side crash and/or rollover events may not be recorded.
- This airbag ECU records post-crash data, and depending on the airbag ECU, may record pre-crash data.
  - If a single event occurs independently, the data for that event is recorded on a one-to-one basis.
  - If multiple events occur successively (within a period of approximately 500ms), the establishment of the recording trigger for the first event is defined as the "pre-crash recording trigger". Pre-crash data for the first event and post-crash data for each successive event is then recorded.
- The airbag ECU has two recording pages (memory maps) to store pre-crash data. Additionally, to store post-crash data, the airbag ECU has two recording pages for each accident type: two pages for frontal and rear crash, two pages for a side crash, and two pages for rollover event.
- The data recorded by the airbag ECU includes correlating information between each previously occurring event (i.e., information that clarifies the collision event sequence. This correlation information consists of the following items.
  - Time from Previous Pre-Crash TRG
  - Linked Pre-Crash Page
  - Time from Pre-Crash TRG

- TRG Count
- Previous Crash Type

- The point in time at which the recording trigger is established is regarded as time zero for the recorded data.
- The recording trigger judgment threshold value differs depending on the collision type (i.e., frontal crash, rear crash, side crash, or rollover event).
- Time series data for side crash may have 24 or 25 sampling points.
- Some of the data recorded by the airbag ECU is transmitted to the airbag ECU from various vehicle control modules by the vehicle's Controller Area Network (CAN).
- In some cases, the airbag ECU part number printed on the ECU label may not match the airbag ECU part number that the CDR tool reports. The part number retrieved by the CDR tool should be considered as the official ECU part number.

Data Element Sign Convention:
The following table provides an explanation of the sign notation for data elements that may be included in this CDR report.

Data Element Name	Positive Sign Notation Indicates
Max. Longitudinal Delta-V	Forward
Longitudinal Delta-V	Forward
Max. Lateral Delta-V , B-Pillar Sensor	Outside to Inside
Max. Lateral Delta-V , C-Pillar Sensor	Outside to Inside
Max. Lateral Delta-V , Slide Door Sensor	Outside to Inside
Lateral Delta-V , B-Pillar Sensor	Outside to Inside
Lateral Delta-V , C-Pillar Sensor	Outside to Inside
Lateral Delta-V , Slide Door Sensor	Outside to Inside
Lateral Delta-V , Airbag ECU Sensor	Outside to Inside
Roll Angle Peak	Clockwise Rotation
Roll Angle	Clockwise Rotation
Lateral Acceleration , Airbag ECU Sensor *	Right to Left

* For sensing a rollover

Data Definitions:

- The "ON" setting for the "Freeze Signal" indicates a state in which the non-volatile memory can not be overwritten or deleted by the airbag ECU. After "Freeze Signal" has been turned ON, subsequent events will not be recorded.
- "Recording Status" indicates a state in which all recorded event data has been written into the non-volatile memory, or a state in which this process was interrupted and not fully written into the non-volatile memory. If "Recording Status" is "Incomplete", recorded event data may not be valid.
- "Time to Deployment Command" indicates the time between recording trigger establishment and the determination of airbag deployment. This value may differ from the actual time it takes for the airbag to fully deploy.
- Even if an airbag/pretensioner did not deploy due to the "front passenger airbag disable switch and/or "RSCA Disable Switch" in the ON position or other disabling criteria are met, the "Time to deployment command" data element for that airbag/pretensioner may still be recorded.
- "Engine RPM" indicates the number of engine revolutions, not the number of motor revolutions. The recorded value has an upper limit of 5,200 rpm. Resolution is 400 rpm and the value is rounded down and recorded. For example, if the actual engine speed is 799 rpm, the recorded value will be 400 rpm.
- The upper limit for the recorded "Vehicle Speed" value is 122 km/h (75.8mph). Resolution is 2km/h (1.2mph) and the value is rounded down and recorded. The accuracy of the "Vehicle Speed" value can be affected by various factors. These include, but not limited, to the following.
  - Significant changes in the tire's rolling radius
  - Wheel lock and wheel slip
- "Accelerator Rate" value is recorded as a voltage. The voltage increases as the driver depresses the accelerator.
- The "Drive" setting for the "Shift Position" value indicates the shift position state is other than "R,"(Reverse), "N" (Neutral), or "P" (Park).
- Depending on the type of occupant sensor installed in the vehicle, one of the following three recording formats for "Occupancy Status, Passenger" will be utilized.
  - Occupied / Not Occupied
  - Adult / Child / Not Occupied
  - AM50 / AF05 / Child / Not Occupied
- Resolution of the "Air Bag Warning Lamp ON Time Since DTC was Set" is 15 minutes, and the value is rounded down and recorded.
- "Longitudinal Delta-V" indicates the change in forward speed after establishment of the recording trigger. This does not refer to vehicle speed, and it does not include the change in speed during the period from the start of the actual collision to establishment of the recording trigger.
- Depending on the specifications, "Roll Angle peak" can be recorded as absolute value.
- "Roll Angle peak" may not always match the peak value within the "Roll Angle" sampling points due to differences in data calculation method.
- For "Lateral Delta-V", the sensor location (B-pillar, front door, C-pillar, and slide door) shows the outline of a typical sensor position. Sensory location can be confirmed using the repair manual.
- "Time from Previous Pre-Crash TRG" indicates the time between the establishment of an event's pre-crash recording trigger to the establishment of a more recent event's pre-crash recording trigger. The upper limit for the recorded value is 16,381 milliseconds. In the event of establishment of the first pre-crash recording trigger after the ignition is switched ON, the upper limit value(max value) is recorded.

- "TRG Count" indicates a calculated value of the number of times recording triggers have been established for all crash types. The sequence in which each event occurred can be verified from the "TRG Count". The smaller the "TRG Count" value, the older the data. The upper limit for the recorded value is 65,533 times. When more than one event reaches the upper limit, the actual "TRG Count" may be greater than what is displayed for that event.
- "Linked Pre-Crash Page" is used to link 'paged" pre-crash data with 'paged" post-crash data. When old pre-crash data is overwritten by new pre-crash data, the "Linked Pre-Crash Page" value may record a page number that is not actually linked.
- Resolution of the "Time from Pre-Crash to TRG" is 100 [ms], and the value is rounded down and recorded.

05004_ToyotaDENSO_r015

System Status at Time of Retrieval

ECU Part Number	89170-42380
ECU Generation	04EDR
Recording Status, All Pages	Complete
Freeze Signal	ON
Freeze Signal Factor	Front Airbag Deployment Front Pretensioner Deployment
Diagnostic Trouble Codes Exist	No
Time from Previous Pre Crash TRG (msec)	16381 or greater
Latest Pre-Crash Page	0
Contains Unlinked Pre-Crash Data	No

Event Record Summary at Retrieval

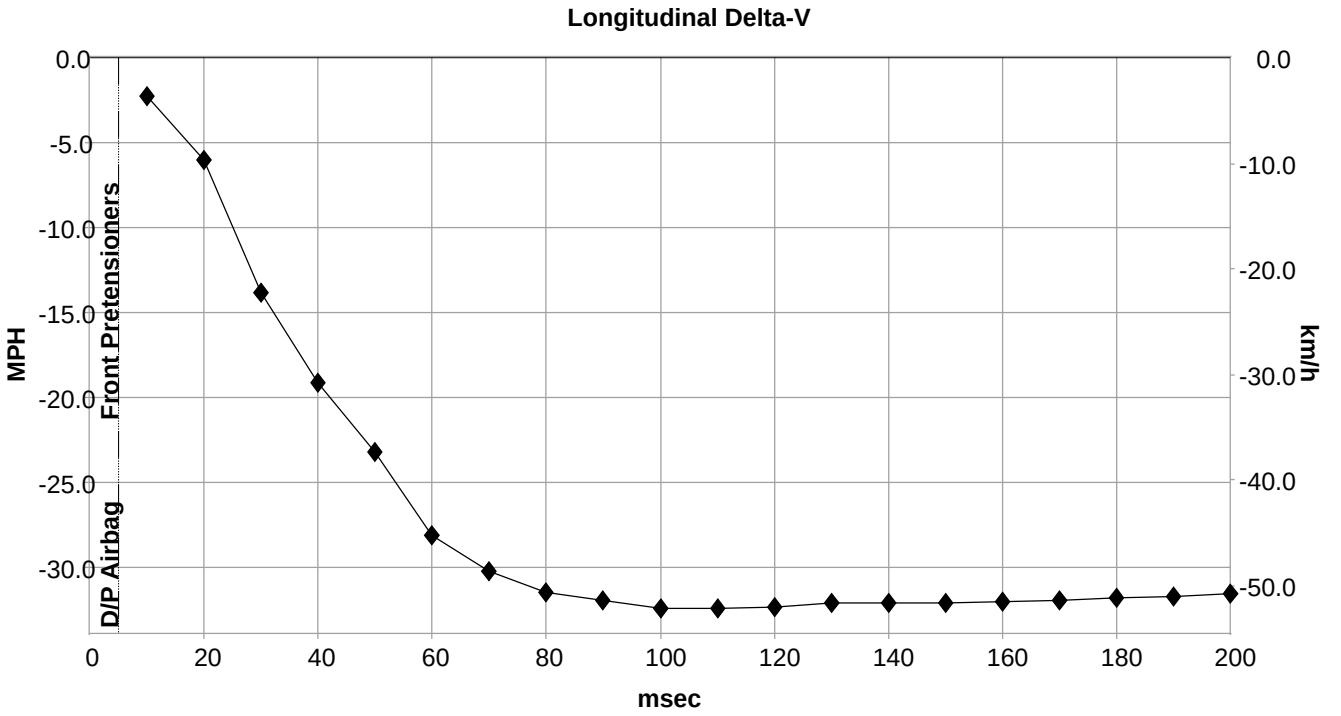
Events Recorded	TRG Count	Crash Type	Time (msec)	Pre-Crash & DTC Data Recording Status	Event & Crash Pulse Data Recording Status
Most Recent Event	2	Front/Rear Crash	0	Complete (Page 0)	Complete (Front/Rear Page 0)
1st Prior Event	1	Side Crash	-1	Complete (Page 0)	Complete (Side Page 0)

System Status at Event (Most Recent Event, TRG 2)

Recording Status, Front/Rear Crash Info.	Complete
Crash Type	Front/Rear Crash
TRG Count (times)	2
Previous Crash Type	Side
Time from Pre-Crash TRG (msec)	1
Linked Pre-Crash Page	0
Time to Deployment Command, Front Airbag, Driver (msec)	5
Time to Deployment Command, Front Airbag, Passenger (msec)	5
Event Severity Status, Driver	Level 3
Event Severity Status, Passenger	N/A
Time to Deployment Command, Pretensioner (msec)	5

Longitudinal Crash Pulse (Most Recent Event, TRG 2 - table 1 of 2)

Recording Status, Time Series Data	Complete
Max Longitudinal Delta-V (MPH [km/h])	-32.4 [-52.1]



Longitudinal Crash Pulse (Most Recent Event, TRG 2 - table 2 of 2)

Time (msec)	Longitudinal Delta-V (MPH [km/h])
10	-2.2 [-3.6]
20	-6.0 [-9.7]
30	-13.8 [-22.2]
40	-19.1 [-30.8]
50	-23.2 [-37.4]
60	-28.1 [-45.2]
70	-30.3 [-48.7]
80	-31.4 [-50.6]
90	-32.0 [-51.4]
100	-32.4 [-52.1]
110	-32.4 [-52.1]
120	-32.3 [-52.0]
130	-32.1 [-51.7]
140	-32.1 [-51.7]
150	-32.1 [-51.7]
160	-32.0 [-51.6]
170	-32.0 [-51.4]
180	-31.8 [-51.2]
190	-31.7 [-51.0]
200	-31.5 [-50.8]

DTCs Present at Time of Event (Most Recent Event, TRG 2)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Airbag Warning Lamp ON Time Since DTC was Set (min)	0
Diagnostic Trouble Codes	None

Pre-Crash Data, 1 Sample (Most Recent Event, TRG 2)

Recording Status, Pre-Crash/Occupant	Complete
Time from Pre-Crash to TRG (msec)	400
Buckle Switch, Driver	Buckled
Buckle Switch, Passenger	Unbuckled
Occupancy Status, Passenger	Not Occupied
Seat Position, Driver	Rearward
Shift Position	Drive

Pre-Crash Data, -5 to 0 seconds (Most Recent Event, TRG 2)

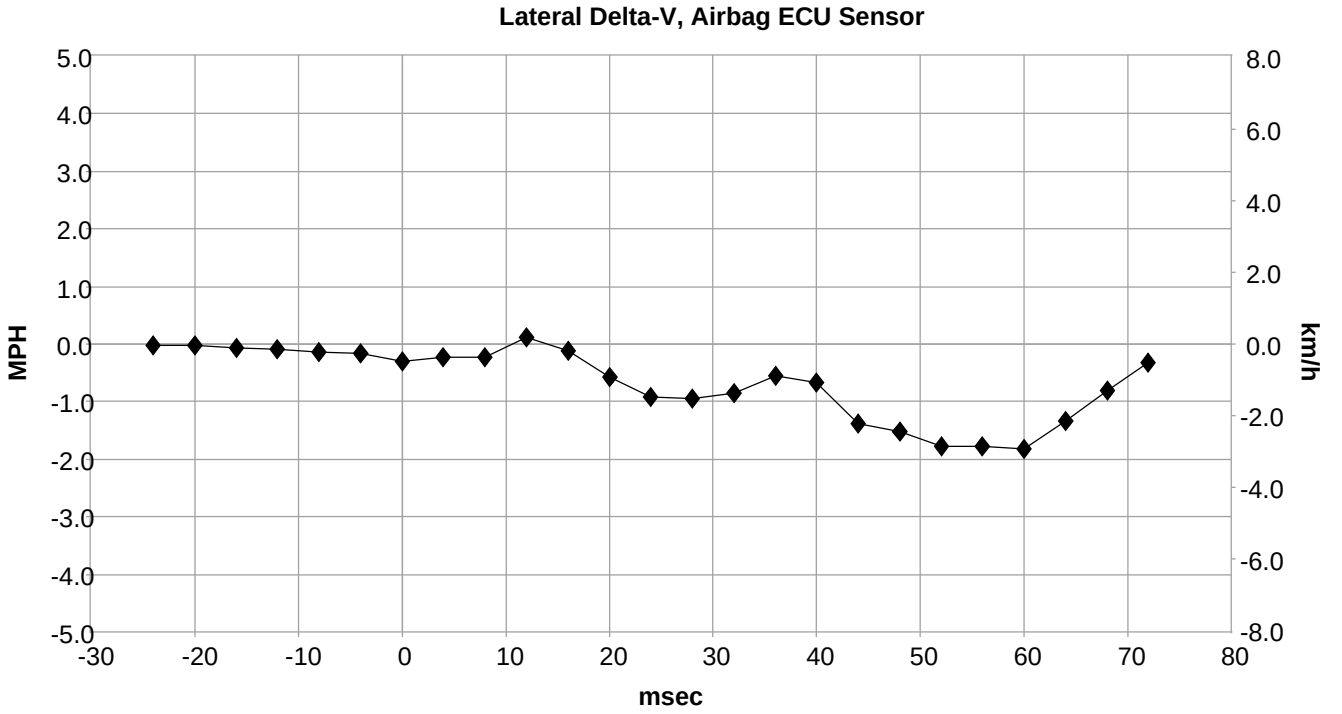
Time (sec)	-4.4	-3.4	-2.4	-1.4	-0.4	0 (TRG)
Vehicle Speed (MPH [km/h])	48.5 [78]	49.7 [80]	49.7 [80]	49.7 [80]	42.3 [68]	32.3 [52]
Brake Switch	OFF	OFF	OFF	ON	ON	ON
Accelerator Rate (V)	0.98	0.86	0.78	0.78	0.78	0.78
Engine RPM (RPM)	1,600	1,600	1,600	1,600	1,200	800

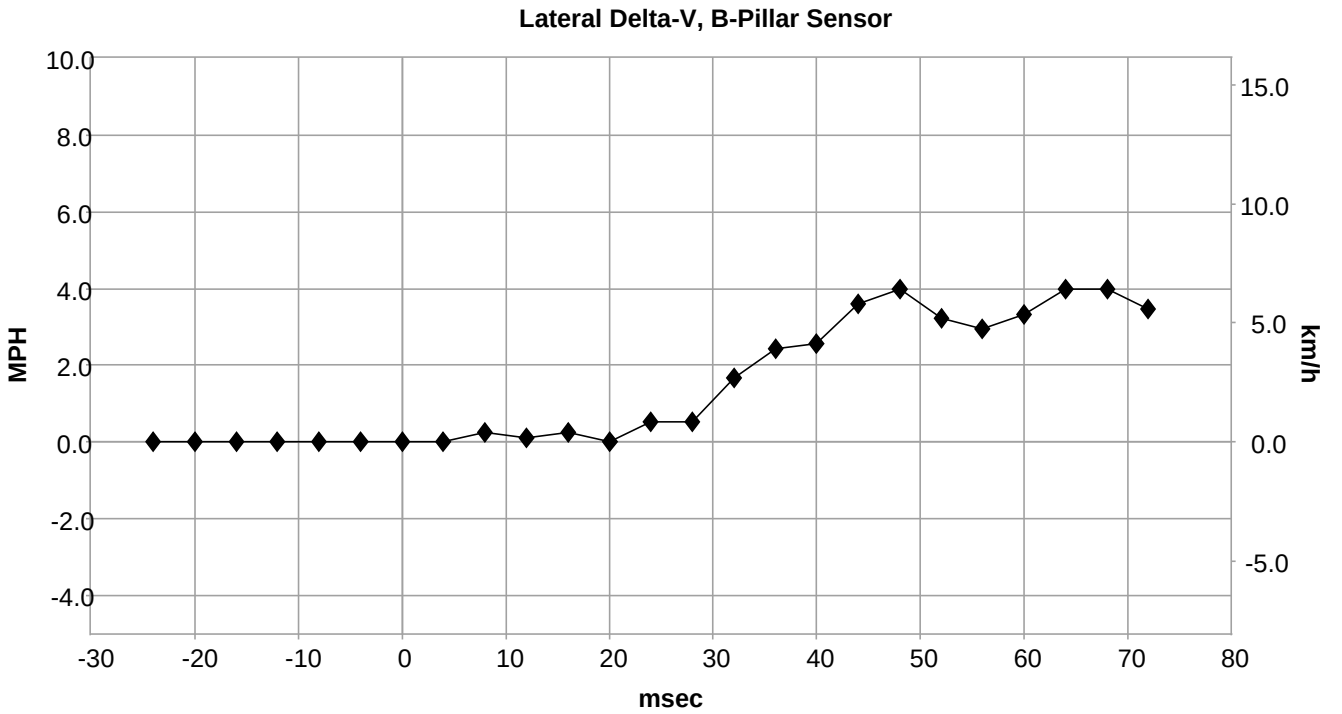
System Status at Event (1st Prior Event, TRG 1)

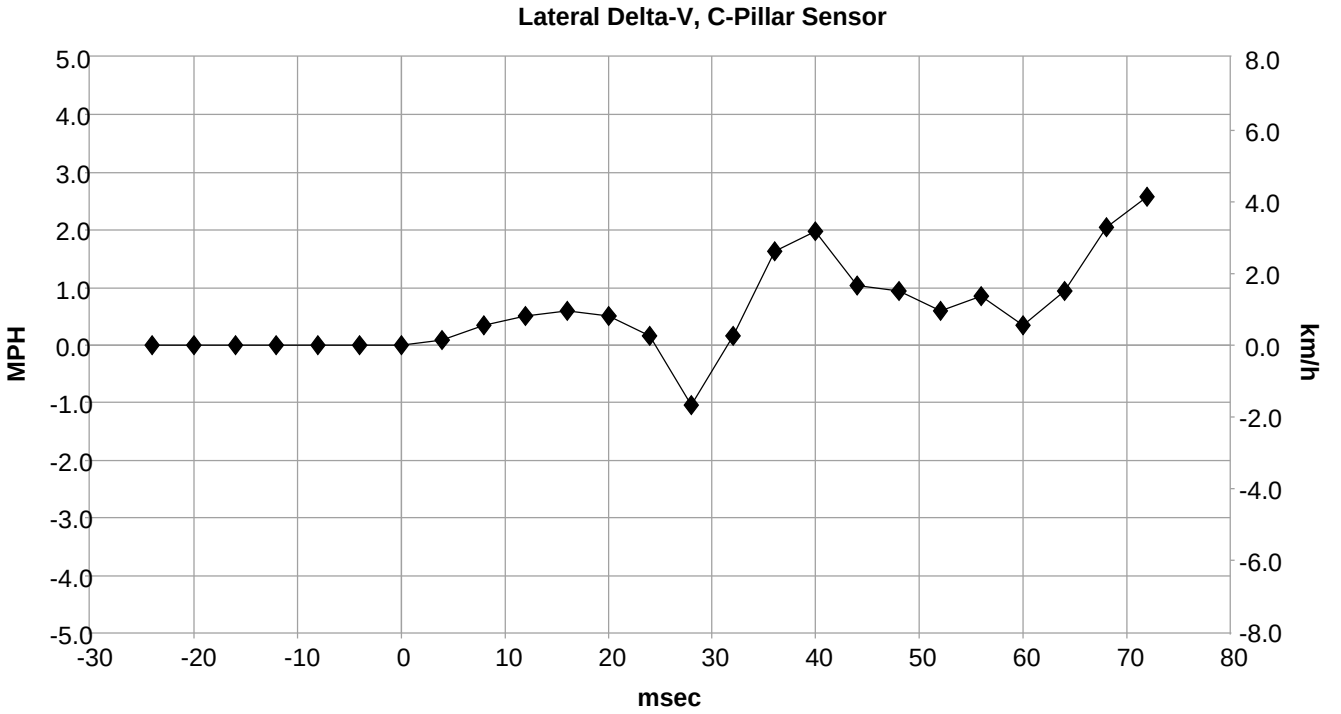
Recording Status, Side Crash Info.	Complete
Crash Type	Side Crash
TRG Count (times)	1
Recorded Side	Passenger's Side
Previous Crash Type	No Event
Time from Pre-Crash TRG (msec)	0
Linked Pre-Crash Page	0
Time to Deployment Command, B-Pillar Sensor (msec)	Not Commanded
Time to Deployment Command, C-Pillar Sensor (msec)	Not Commanded

Lateral Crash Pulse (1st Prior Event, TRG 1 - table 1 of 2)

Recording Status, Time Series Data	Complete
Time from TRG to Next Sample (msec)	0
Max Lateral Delta-V, B-Pillar Sensor (MPH [km/h])	4.0 [6.4]
Max Lateral Delta-V, C-Pillar Sensor (MPH [km/h])	2.6 [4.1]






Lateral Delta-V, C-Pillar Sensor
MPH
km/h
msec
5.0
4.0
3.0
2.0
1.0
0.0
-1.0
-2.0
-3.0
-4.0
-5.0
8.0
6.0
4.0
2.0
0.0
-2.0
-4.0
-6.0
-8.0
-30
-20
-10
0
10
20
30
40
50
60
70
80

Lateral Crash Pulse (1st Prior Event, TRG 1 - table 2 of 2)

Time (msec)	Lateral Delta-V, Airbag ECU Sensor (MPH [km/h])	Lateral Delta-V, B-Pillar Sensor (MPH [km/h])	Lateral Delta-V, C-Pillar Sensor (MPH [km/h])
-24	0.0 [0.0]	0.0 [0.0]	0.0 [0.0]
-20	0.0 [-0.1]	0.0 [0.0]	0.0 [0.0]
-16	-0.1 [-0.1]	0.0 [0.0]	0.0 [0.0]
-12	-0.1 [-0.2]	0.0 [0.0]	0.0 [0.0]
-8	-0.1 [-0.2]	0.0 [0.0]	0.0 [0.0]
-4	-0.2 [-0.3]	0.0 [0.0]	0.0 [0.0]
0	-0.3 [-0.5]	0.0 [0.0]	0.0 [0.0]
4	-0.2 [-0.4]	0.0 [0.0]	0.1 [0.1]
8	-0.2 [-0.4]	0.3 [0.4]	0.3 [0.6]
12	0.1 [0.2]	0.1 [0.2]	0.5 [0.8]
16	-0.1 [-0.2]	0.3 [0.4]	0.6 [1.0]
20	-0.6 [-0.9]	0.0 [0.0]	0.5 [0.8]
24	-0.9 [-1.5]	0.5 [0.8]	0.2 [0.3]
28	-0.9 [-1.5]	0.5 [0.8]	-1.0 [-1.7]
32	-0.9 [-1.4]	1.7 [2.7]	0.2 [0.3]
36	-0.5 [-0.9]	2.4 [3.9]	1.6 [2.6]
40	-0.7 [-1.1]	2.6 [4.1]	2.0 [3.2]
44	-1.4 [-2.2]	3.6 [5.8]	1.0 [1.7]
48	-1.5 [-2.5]	4.0 [6.4]	0.9 [1.5]
52	-1.8 [-2.8]	3.2 [5.2]	0.6 [1.0]
56	-1.8 [-2.9]	3.0 [4.8]	0.9 [1.4]
60	-1.8 [-2.9]	3.3 [5.4]	0.3 [0.6]
64	-1.3 [-2.2]	4.0 [6.4]	0.9 [1.5]
68	-0.8 [-1.3]	4.0 [6.4]	2.1 [3.3]
72	-0.3 [-0.5]	3.5 [5.6]	2.6 [4.1]

DTCs Present at Time of Event (1st Prior Event, TRG 1)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Airbag Warning Lamp ON Time Since DTC was Set (min)	0
Diagnostic Trouble Codes	None

Pre-Crash Data, 1 Sample (1st Prior Event, TRG 1)

Recording Status, Pre-Crash/Occupant	Complete
Time from Pre-Crash to TRG (msec)	400
Buckle Switch, Driver	Buckled
Buckle Switch, Passenger	Unbuckled
Occupancy Status, Passenger	Not Occupied
Seat Position, Driver	Rearward
Shift Position	Drive

Pre-Crash Data, -5 to 0 seconds (1st Prior Event, TRG 1)

Time (sec)	-4.4	-3.4	-2.4	-1.4	-0.4	0 (TRG)
Vehicle Speed (MPH [km/h])	48.5 [78]	49.7 [80]	49.7 [80]	49.7 [80]	42.3 [68]	32.3 [52]
Brake Switch	OFF	OFF	OFF	ON	ON	ON
Accelerator Rate (V)	0.98	0.86	0.78	0.78	0.78	0.78
Engine RPM (RPM)	1,600	1,600	1,600	1,600	1,200	800